SEP490 · CAPSTONE PROJECT · REVIEW 2 · BM01

Design Brief – Review 2

0. Header

Project code	FA26SE905
Project name	Intelligent Hotel Concierge Robot (HC-Robot)
Team	GFA26SE136
Supervisor	NhanDT35
Repository link (tag review-2)	https://github.com/HC-Robot-Smart-Concierge/HCROBOT.git
Version	2.0
Date	9/11/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
Core Workflows	Fragmented single-action flows (MOVE, GREET, FEEDBACK) centered solely on robot.	5 standardized workflows: Admin Configuration, Guest Info & Recommendation, Service Request Handling, Robot-to-Human Handoff, and Robot Monitoring & Incident Handling.	Replace micro-actions with comprehensive multi-actor SOPs covering guest service, administration, and incident recovery.
Robot Monitoring & Telemetry	Basic status indicator without centralized telemetry.	Real-time monitoring tracking Cartesian pose (x, y, yaw), velocity, LiDAR scans, and hardware health (COM/ultrasonic/Pi 5).	Fulfill supervisor requirement for live fleet localization and device health tracking.

Section	Current	Proposed	Reason
Battery Buffer Strategy	Single low-battery trigger without operational buffers.	Dual-buffer policy: 20% Warning Buffer (drain current task, show banner, block new sessions), 10% Critical Cut-off (abort task, alert modal, auto-dock, alert maintenance), and 80% resume hysteresis.	Fulfill supervisor requirement for explicit buffer thresholds for guest notices, UI warnings, and auto-docking.

1.2 Out of Scope

Item	Reason
Restaurant Dine-In & Table POS	System handles Hotel Kitchen room-service ticket dispatch (kitchen.py); on-premise table seating and dine-in are excluded.
Multi-Floor Elevator Integration	Hardware navigation is restricted to single-floor lobby, lounge, and reception areas for Review 2 defense.
Physical Luggage Transportation	Robot scope is information and escort concierge; heavy luggage is handled by bell staff.
Autonomous Outdoor Navigation	LiDAR and ultrasonic safety ranges are calibrated exclusively for indoor hotel flooring.

2. System at a Glance

The Intelligent Hotel Concierge Robot (HC-Robot) resolves front-desk congestion, repetitive guest inquiries, and delayed service routing in luxury hospitality. Designed for Hotel Guests, Front Desk Staff, Maintenance Operators, and Hotel Administrators, HC-Robot delivers bilingual voice AI interactions (Groq/Gemini LLM with local RAG factsheets and EdgeTTS), autonomous visual escorting using 2D LiDAR SLAM, and automated hotel service request dispatching. All operations are backed by a real-time fleet telemetry dashboard tracking pose coordinates, device sensor health, and a fail-safe multi-tier battery buffer architecture.

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	Multi-Tier Hybrid AI Voice Pipeline	(A) Local Ollama; (B) Cloud LLM; (C) Tier 1 Groq / Tier 2 Gemini + Local RAG factsheets + EdgeTTS.	Option C. Reduces conversational latency from 15s to <2s on edge hardware with offline factsheet fallback.	Kha	FR-AI-01 / Architecture Fig 2.1
D2	Dual-Threshold Battery Buffer & Docking	(A) Single 15% threshold; (B) Dual buffer (20% warning drain, 10% critical cut-off, 80% recharge hysteresis).	Option B. Prevents mid-lobby stalls while avoiding abrupt session termination at safe battery levels.	Hưng	FR-PWR-02 / Battery State Machine
D3	2 Hz Telemetry & Hardware Health Stream	(A) 5s REST polling; (B) 2-5 Hz WebSocket broadcasting pose, LiDAR scans, velocities, and hardware bus health.	Option B. Enables zero-latency map tracking and immediate detection of sensor/COM disconnects.	Phúc	FR-MON-01 / Telemetry Flowchart
D4	Standardized Core Workflows (WF-01..05)	(A) 10 micro-actions; (B) 5 standardized operational workflows (Admin Config, Guest Info, Request Handling, Handoff, Monitoring).	Option B. Establishes realistic operational procedures directly aligned with presentation slides and defense targets.	Vinh	FR-WF-01..05 / Core Workflows: • Admin configuration • Guest information & Recommendation • Guest Service Request Handling • Robot-to-Human Handoff • Robot Monitoring & Incident Handling
D5	Robot-to-Human Handoff with Translation	(A) Text ticket; (B) 2-way WebRTC screen call with live speech translation (EN, VN) triggered by request or bad expression.	Option B. Enables instant front-desk takeover when guest requests staff or displays frustration.	Kiệt	FR-CALL-01 / Handoff Architecture

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Battery exhaustion during guest escort.	Robot stalls mid-corridor, blocking lobby passage.	20% warning blocks new tasks; 10% critical cut-off aborts task, plays voice apology, auto-docks, and alerts maintenance.	Phúc
R2	Sensor disconnect (LiDAR COM port / ultrasonic).	Blind navigation, collision hazard, lost pose.	Watchdog halts motors (PWM=0, EMERGENCY_STOP) if packets drop >1.0s; triggers red alert on Admin Portal.	Hưng
R3	High LLM latency or cloud API outage.	Dialogue stalls with dead silence.	Earcon turn chime, visual thinking indicator, barge-in voice interruption, and auto-fallback to local cached factsheets.	Kha

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G-01	Battery buffer logic in frontend was unaligned with backend robot model.	frontend/.../batteryPolicy.js vs backend/app/models/robot.py	Align Robot.status flags with 20% LOW and 10% CRITICAL thresholds; verify via integration test.	Hưng	2026-11-12
G-02	Fleet telemetry stream lacked structured hardware health metrics.	backend/app/api/v1/endpoints/map.py	Add structured device_health (LiDAR COM, ultrasonic clearances) to WebSocket stream.	Vinh	2026-11-13

ID	Gap	Location	Plan	Owner	Due
G-03	robot UI lacked persistent warning banner in 10-20% buffer zone.	frontend/src/pages /guest/RobotScreen Page.jsx	Implement top warning notification bar informing guests of upcoming recharge return.	Kha	2026-11-14

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Admin Knowledge & Configuration	Hotel Admin	Admin logs in, updates knowledge files, FAQs, promotions, and expressions, syncs to robot, and verifies deployment in < 3.0s.	TC-F01-01..03	Done
F-02	Guest Info & Recommendation	Hotel Guest	Detects guest approach, prompts language, answers queries via AI voice in < 2.0s, provides recommendations, and collects rating.	TC-F02-01..03	Done
F-03	Service Request Handling & Routing	Hotel Guest / Head Lead / Staff	Classifies service requests from robot, routes ticket to target department, and updates fulfillment status to guest in real time.	TC-F03-01..03	Done
F-04	Robot-to-Human Handoff & Translation	Hotel Guest / Front Desk Staff	Triggers on guest request or bad expression detection; connects 2-way screen call with live EN-VN translation.	TC-F04-01..03	Done

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-05	Robot Monitoring & Incident Handling	Operations Staff / Admin	Tracks live pose, velocity, battery %, and LiDAR scans at $\geq$ 2 Hz; engages emergency stop and alerts staff upon anomaly.	TC-F05-01..03	Done
F-06	Multi-Tier Battery Buffer & Auto-Docking	Concierge Robot / Maintenance Staff	Blocks new sessions and displays banner at $\leq$ 20%; aborts session, shows modal, docks, and creates ticket at $\leq$ 10%; resumes at $\geq$ 80%.	TC-F06-01..04	In progress

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	Performance (Voice Latency)	End-to-end voice query response (STT + LLM RAG + TTS)	$\leq$ 2.0 seconds	Timestamp delta between speech end and audio playback start across 50 sample queries.
NFR-02	Safety (Battery Buffer)	Autonomous docking before complete power loss	0 mid-lobby stalls; auto-dock triggers at $\geq$ 10%	Automated discharge stress test repeating escort runs until 20% and 10% triggers.
NFR-03	Telemetry Freshness	WebSocket update frequency and latency to Admin	$\geq$ 2 Hz; latency $\leq$ 500 ms	Client-side packet arrival timestamp delta during active navigation.
NFR-04	Navigation Safety	Collision avoidance emergency stop distance	Stop distance $\leq$ 0.3 meters at 0.35 m/s	Physical sudden-obstacle insertion in trajectory; ultrasonic and LiDAR reaction test.
NFR-05	System Reliability	Service uptime during lobby operating hours (07:00 - 22:00)	$\geq$ 99.0% uptime	Server heartbeat logs and error rate tracking over 7 consecutive operational days.
NFR-06	Knowledge Sync Latency	Admin configuration to robot active knowledge sync	$\leq$ 3.0 seconds	Timestamp delta between admin save and robot RAG vector index reload.

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Phúc	Fullstack Developer (Leader)	Admin Portal, Expression / Promotion / FAQ Config, Knowledge Base Management	WF-01: Admin Configuration	D4 (Standardized Workflows), PR #31 (Admin Config & CI Test Suite)
Hưng	Fullstack Developer	Human Approach Detection, Voice Recommendation, Hotel Info / Instructions	WF-02: Guest Information & Recommendation	D1 (Hybrid AI Voice Pipeline), PR #22 (Groq/Gemini Multi-Tier Engine)
Kha	Fullstack Developer	Department Ticket Routing, Room Service Dispatch, Head Lead Approval Flow	WF-03: Guest Service Request Handling	D4 (Service Dispatch), PR #15 (Ticketing & Notification System)
Kiệt	Fullstack Developer	robot Touchscreen, 2-Way Screen Video Call, Live Speech Translation (EN, VN)	WF-04: Robot-to-Human Handoff	D5 (Handoff & Video Translation), PR #27 (Video Call & Audio Stream)
Vinh	Fullstack Developer	ROS2 Navigation, Fleet Telemetry, LiDAR Watchdog & Battery Buffer Docking	WF-05: Robot Monitoring & Incident Handling	D2 (Battery Policy), D3 (Telemetry WebSocket), PR #18 (Fleet Monitor)

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Admin Configuration	Hotel Admin, Backend, Robot	Admin Login on management portal	Login → Update knowledge base, FAQ, promotions, expressions → Synchronize → Verify deployment on robot.	F-01	AdminSession: AUTHENTICATED; KnowledgeBase.status: DRAFT → PUBLISHED;Robot: SYNCED	Phúc
WF-02	Guest Information & Recommendation	Hotel Guest, Concierge Robot	Human approach detection (camera / touchscreen tap)	Detect guest → Greet & select language → Q&A, service recommendations → Collect guest rating/feedback.	F-02	Robot.status: IDLE → INTERACTING → IDLE;Feedback.rating: recorded	Kha
WF-03	Guest Service Request Handling	Hotel Guest, Robot, Department Staff, Head Lead	Guest submits service request (Room Service, Taxi, Housekeeping)	Guest submits request → Classify & route to target department → Staff receives & fulfills → Update status to guest.	F-03	SupportRequest.status: NEW → ASSIGNED → IN_PROGRESS → COMPLETED	Vinh
WF-04	Robot-to-Human Handoff	Hotel Guest, Concierge Robot, Front Desk Staff	Guest requests staff OR robot detects bad expression	Guest requests assistance → Notify staff → Staff answers call → Communicate through robot screen with translation → End session.	F-04	ChatSession.status: AI_ACTIVE → ESCALATED_HUMAN; SupportRequest.status: PENDING → RESOLVED	Kiệt

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-05	Robot Monitoring & Incident Handling	Operations Staff, Maintenance, Robot, Backend	Periodic 2 Hz telemetry stream OR detected anomaly (LiDAR drop / low battery)	Monitor status → Detect anomaly (battery $\leq 10\%$, LiDAR drop) → Alert staff → Staff handles incident → Update status and log history.	F-05, F-06	Robot.status: IDLE / NAVIGATING → EMERGENCY_STOP / CHARGING; IncidentLog.status: LOGGED → RESOLVED	Hưng

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	WF-01	Main	Admin logs into portal; new promotion and FAQ items prepared.	1. Admin logs into management portal.2. Updates FAQ entries, promotions, and robot expressions.3. Clicks "Synchronize to Robot".4. Verifies updated content on robot robot screen.	New FAQs and promotions immediately synced and rendered on robot without system downtime.	Phúc
DS-02	WF-01	Exception	Admin uploads unsupported or corrupted document file.	1. Admin uploads invalid document file.2. System validates and rejects document.3. Validation error displayed; sync aborted.4. Robot maintains existing valid knowledge base.	Rejection alert shown; robot continues running on previous stable knowledge base.	Phúc
DS-03	WF-02	Main	Robot idle in lobby; guest approaches robot.	1. Human approach detection triggers.2. Robot greets guest and prompts language selection.3. Guest asks questions and receives amenity recommendations.4. Guest submits 5-star rating on screen.	Smooth conversational flow, personalized recommendations displayed, rating persisted.	Phúc

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04	WF-02	Exception	Guest walks away mid-conversation without rating (15s inactivity).	1. Interaction active on screen.2. Guest walks away without completing survey.3. Inactivity timer reaches 15s without touch/voice.4. Session auto-closes and resets to idle screensaver.	Session auto-recovers cleanly without getting stuck on active conversation.	Phúc
DS-05	WF-03	Main	Guest submits room service request (Room 402) on robot.	1. Guest selects service request on robot.2. System classifies request and routes to target department.3. Department staff receives and accepts request.4. Head Lead confirms and updates status to completed.	Ticket lifecycle completes (NEW → ASSIGNED → COMPLETED); guest notified on screen.	Kha
DS-06	WF-03	Exception	Guest requests service during department off-hours.	1. Guest submits off-hours service request.2. System detects target department unavailable.3. Robot prompts scheduled morning booking.4. Ticket queued with scheduled status.	Clear instructions given to guest; ticket properly scheduled without being lost.	Kha
DS-07	WF-04	Main	Guest during normal communication asks to speak with staff.	1. Guest speaks: "I want to talk to front desk".2. Robot alerts staff & displays waiting screen.3. Staff picks up call; screen communication connects.4. Live speech translation active (English, VN).	Successful 2-way screen communication with synchronized bilingual subtitles.	Kha

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-08	WF-04	Exception	Guest displays frustration during interaction.	1. Normal communication in progress.2. Robot detects guest's bad expression.3. Robot automatically alerts staff.4. Staff picks up call to assist guest directly.	Automated escalation without requiring guest to manually find or tap help buttons.	Kha
DS-09	WF-05	Main	Operations staff monitors fleet on Admin Portal.	1. Open Admin Physical Map tab.2. Observe live pose and 360° LiDAR scan overlay.3. Verify hardware indicators (Battery, LiDAR, Ultrasonic, Speed).	Robot marker moves synchronously at ≥ 2 Hz; all hardware status pills display nominal green.	Kha
DS-10	WF-05	Exception	Sensor disconnect (LiDAR COM port) or critical battery ($\leq 10\%$).	1. Sensor packet drop > 1.0s or battery hits 10%.2. Watchdog triggers EMERGENCY_STOP / auto-dock.3. Red alert banner and audio chime fire on portal.4. Staff receives alert, resolves incident, updates log.	Emergency halt/docking executes safely; incident logged in audit history.	Hưng

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	6	3	18
Main workflows (end-to-end)	5	2	10
State transitions of main entities	5	2	10
Agreed NFRs	6	2	12
Committed total	22		50

Automated in CI: at least 50% of the committed total (26/50 automated test cases via Pytest & Vitest in GitHub Actions).